Antonio SILVETI-FALLS

Personal Data

	American | Currently based in Paris, France | Born 09 October 1992
email:	TonySFalls@gmail.com
Languages:	English (native), French (C1)
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Research Interests

Optimization theory and deep learning; geometry-aware methods for scalable pretraining; stochastic and nonsmooth optimization; implicit differentiation, conservative field theory, and o-minimal geometry for neural networks with implicitly defined layers.

Current Position (Sept 2022 -)

Maître de conférences (Assistant/associate professor) at CentraleSupélec/Université de Paris-Saclay in the Centre pour la Vision Numérique (CVN) Laboratory (assoc. inria OPIS Team and Fédération de Mathématiques de CentraleSupélec).

Research Program

My research lies at the interface of rigorous optimization theory and deep learning practice. It is organized around two complementary strands: (i) geometry-aware optimization methods for scalable pretraining of transformers and large models, including training dynamics, scaling laws, clipping and stability, and transfer across model scales and architectures; and (ii) nonsmooth and o-minimal geometric frameworks for extending backpropagation and implicit differentiation to neural networks with implicitly defined layers, with rigorous convergence guarantees via conservative field theory, definable and semialgebraic functions, and KL inequalities.

Grants and Research Impact

- PI of the ANR JCJC grant project "SIMPLES" (241k €, 4 years), awarded in August 2025.
- Co-PI with Evgenii Chzhen of a DataIA Modular Chair grant (260k €, 4 years), awarded in February 2026.
- The *Scion* optimizer code arising from this line of work was adopted into NVIDIA NeMo's Emerging-Optimizers codebase.

Ph.D. Thesis (Oct 2017 - Feb 2021)

Title	First-order Noneuclidean Splitting Methods for Large-scale Optimization: Deterministic and Stochastic Algorithms
University	Université de Caen Normandie
Advisors	Jalal Fadili (UniCaen, ENSICAEN) and Gabriel Peyré (ENS Paris, CNRS)

Education

Aug 2015 - June 2017	M.Sc. in Applied Mathematics - Nonlinear Dynamical Systems (GPA: 3.87), San Diego State University, USA. Adviser: Jérôme Gilles. (Thesis: Empirical Gabor Frames)
Aug 2010 - June 2015	B.Sc. in Mathematics, California State University - Chico, USA. Adviser: Thomas Mattman.
Aug 2010 - June 2015	B.Sc. in Applied Mathematics, California State University - Chico, USA. Adviser: Vladimir Rosenhaus.
Aug 2010 - June 2015	B.Sc. in Statistics, California State University - Chico, USA. Adviser: Kathy Gray.

Teaching Experience

2023 - now — **CentraleSupélec**
Created and taught the course Optimization for Computer Vision as part of the masters degree in AI. Lecturer for the course Convergence-Integration-Probability. Led the "travaux dirigés" for numerous courses: Convergence-Integration-Probability, PDEs, Optimization, Analysis I, III, and IV among many others.

2022 - 2023 — **CentraleSupélec**
Led the "travaux dirigés" for three courses: Convergence-Integration-Probability, PDEs, and Optimization.

2022 — **Toulouse School of Economics**
Led the "travaux dirigés" associated to the course Optimization for Big Data for master's level (M1) students using Python.

2021 — **Toulouse Business School**
Taught 2 sections of Business Data Analysis for master's level (M1) students using R and RStudio.

2015-2017 — **San Diego State University**
Taught 3 semesters (2 sections per semester) of introductory differential calculus.

Publications

15 2026 Cesare Molinari, Antonio Silveti-Falls, Zev Woodstock, *"Frank-Wolfe with Moreau Envelope Smoothing for Nonsmooth Nonconvex Problems"* (in preparation).

14 2026 Navil Manunandhan, Abbas Khademi, Antonio Silveti-Falls, *"Stochastic Boosted Frank-Wolfe"* NeurIPS 2026 (submitted).

13 2026 Konstantinos Oikonomidis, Jan Quan, Kimon Antonakopoulos, Antonio Silveti-Falls, Volkan Cevher, Panagiotis Patrinos, *"Constrained Stochastic Spectral Preconditioning Converges for Nonconvex Objectives"* arXiv preprint arXiv:2605.11850.

12 2026 Rustem Islamov, Roman Machacek, Aurelien Lucchi, Antonio Silveti-Falls, Eduard Gorbunov, Volkan Cevher, *"On the Role of Batch Size in Stochastic Conditional Gradient Methods"* ICML 2026.

11 2025 Thomas Pethick, Kimon Antonakopoulos, Antonio Silveti-Falls, Leena Chennuru Vankadara, Volkan Cevher, *"Training Neural Networks at Any Scale"* arXiv preprint arXiv:2511.11163.

10 2025 Abbas Khademi, Antonio Silveti-Falls, *"Adaptive Conditional Gradient Descent"* (submitted).

9 2025 Miguel Amorim, Jean-Christophe Pesquet, Tristan Portugues, Antonio Silveti-Falls, Hugues Talbot, *"Numerical Implementation of Variational Morphological Operators on Graphs"* DGMM 2025.

8 2025 Thomas Pethick, Wanyun Xie, Mete Erdogan, Kimon Antonakopoulos, Tony Silveti-Falls, Volkan Cevher, *"Generalized Gradient Norm Clipping and Non-Euclidean (L_0, L_1)-smoothness"* NeurIPS 2025 **Oral at NeurIPS 2025**.

7 2025 Thomas Pethick, Wanyun Xie, Kimon Antonakopoulos, Zhenyu Zhu, Antonio Silveti-Falls, Volkan Cevher, *"Training Deep Learning Models with Norm-Constrained LMOs"* ICML 2025 **Spotlight at ICML 2025**.

Publications (Continued)

6	2024	Jérôme Bolte, Edouard Pauwels, Antonio Silveti-Falls, *"Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems"* SIAM Journal on Optimization, Vol. 34, No. 1, pp. 71-97, 2024.
5	2023	Antonio Silveti-Falls, Cesare Molinari, Jalal Fadili, *"An Inexact Bregman Primal-Dual Splitting Algorithm for Composite Optimization"* Pure and Applied Functional Analysis, Vol. 8, No. 3, pp. 921-964, 2023.
4	2021	Jérôme Bolte, Tâm Lê, Edouard Pauwels, Antonio Silveti-Falls, *"Nonsmooth Implicit Differentiation for Machine Learning and Optimization"* Proceedings of the 35th International Conference on Neural Information Processing Systems, 2021.
3	2021	Antonio Silveti-Falls, Cesare Molinari, Jalal Fadili, *"Inexact and Stochastic Generalized Conditional Gradient with Augmented Lagrangian and Proximal Step"* Journal of Nonsmooth Analysis and Optimization, Vol. 2, 2021.
2	2020	Antonio Silveti-Falls, Cesare Molinari, Jalal Fadili, *"Generalized Conditional Gradient with Augmented Lagrangian for Composite Minimization"* SIAM Journal on Optimization, Vol. 30, No. 4, pp. 2687-2725, 2020 **Best student paper at SPARS 2019**.
1	2015	Kathy Gray, Brittany Hampton, Tony Silveti-Falls, Allison McConnel, Casey Bausell, *"Comparison of Bayesian Credible Intervals to Frequentist Confidence Intervals"* Journal of Modern Applied Statistical Methods, Vol. 14, No. 1, pp. 43-52, 2015.

Editorial Activity (Reviewing, Chairing)

- SIAM Journal on Optimization (SIOPT)
- SIAM Journal on Mathematics and Data Science (SIMODS)
- Mathematical Programming
- Journal of Machine Learning Research (JMLR)
- Conference on Neural Information Processing Systems (NeurIPS) 2022 **(Highlighted reviewer)**, 2024 **(Area Chair)**, 2025 **(Area Chair)**, 2026 **(Area Chair)**
- International Conference on Learning Representations (ICLR) 2022 **(Highlighted reviewer)**, 2023, 2024, 2025
- International Conference on Artificial Intelligence and Statistics (AISTATS) 2022, 2023, 2025, 2026
- International Conference on Machine Learning (ICML) 2023, 2024, 2025 **(Area Chair)**, 2026 **(Area Chair)**
- Transactions on Machine Learning Research (TMLR) 2024, 2025
- Journal of Optimization Theory and Applications (JOTA)
- Computational Optimization and Applications (COAP)
- Journal of Scientific Computing
- British Machine Vision Conference 2022
- Montréal AI Symposium 2022

Academic Service

- Advised Navilarasu Subrahmanyan Manunandhan 2023-2025 (Undergraduate research on Stochastic Frank-Wolfe methods)
- Advised Miguel Amorim 2024 (Masters internship on variational mathematical morphology)
- Advised Tristan Portugues 2025 (Masters internship + research stay on variational mathematical morphology)
- Advised Eline Pot 2025-2028 (PhD on Uncertainty Quantification for Neural Networks)
- Thesis reporter for J. Chirinos Rodriguez 2024
- Thesis committee member for Silvia Sciutto 2024
- Masters thesis committee member for Sanberk Serbest (EPFL) 2025

Conference and Seminar Talks

2026	ELLIIT Focus Period Symposium: Optimization for Learning, Lund, Sweden - *Spectral optimizers for deep learning: muon, scion, and so on.*
2026	ENS Ulm, Paris, France - *Training Neural Networks at Any Scale.*
2026	Huawei Lagrange Centre for Mathematics, Paris, France - *Stochastic Conditional Gradient with Operator Norms for Deep Learning.*
2025	MOP Research Retreat, Saarbrucken, Germany - *Training Neural Networks at Any Scale.*
2025	Cohere Labs ML Theory, online - *Training Neural Networks at Any Scale. (YouTube)*
2025	CRM Workshop on optimization and learning: theory and applications, Montreal, Canada - *Training Deep Learning Models with Norm-Constrained LMOs.*
2025	FNRS Contact Group on "Wavelets and Applications", Louvain-la-Neuve, Belgium - *Training Deep Learning Models with Norm-Constrained LMOs.*
2025	Le Séminaire Palaisien, Palaiseau, France - *Training Deep Learning Models with Norm-Constrained LMOs.*
2025	GDR IASIS sur l'optimisation bi-niveau et l'apprentissage des hyperparamètres, Lyon, France - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2025	LPSM Séminaire de Statistique, Paris, France - *Training Deep Learning Models with Norm-Constrained LMOs.*
2025	MOP Seminar, Online - *Training Deep Learning Models with Norm-Constrained LMOs.*
2024	MOP Research Retreat, Saarbrucken, Germany - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2024	TraDE-OPT Final Conference, Sestri Levante, Italy - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2024	Optimization Techniques for Inverse Problems, Modena, Italy - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2024	International Symposium on Mathematical Programming (ISMP), Montreal, Canada - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2024	European Conference on Operational Research (EURO), Technical University Denmark, Copenhagen, Denmark - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2024	Fondements Mathématiques de l'IA, Institut DataIA, Paris, France - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*
2024	CRM Seminar, McGill University, Montreal, Canada - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems.*

Conference and Seminar Talks (Continued)

2023	International Congress on Industrial and Applied Mathematics, Tokyo, Japan - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems*.
2023	ENS Paris Center for Data Science, Paris, France - *Differentiating Nonsmooth Solutions to Parametric Monotone Inclusion Problems*.
2023	SIAM Conference on Optimization, Seattle, Washington - *A Stochastic Bregman Primal-Dual Splitting Algorithm for Composite Optimization*.
2023	ENS Lyon Machine Learning Seminar, Lyon, France - Nonsmooth implicit differentiation.
2022	GREYC Lab Seminar Caen, France - *Nonsmooth Implicit Differentiation for Machine Learning and Optimization*.
2022	Inauguration de la Fédération de Mathématiques de CentraleSupélec - *Nonsmooth Implicit Differentiation for Machine Learning and Optimization*.
2022	University of Tubingen MOP Research Seminar (virtual) - *Nonsmooth Implicit Differentiation for Machine Learning and Optimization*.
2022	Curves and Surfaces Arcachon - *Nonsmooth Implicit Differentiation for Machine Learning*.
2022	University of Basel Department of Mathematics and Computer Science Seminar - *Nonsmooth Implicit Differentiation for Machine Learning and Optimization*.
2022	Centre de Vision Numérique in Gif-sur-Yvette, France - *Nonsmooth Implicit Differentiation for Machine Learning and Optimization*.
2022	Institute of Mathematics and Scientific Computing at Graz, Austria - *Nonsmooth Implicit Differentiation for Machine Learning*.
2022	Séminaire Modélisation, Optimisation, Dynamique Université de Limoges, France - *Différentiation Implicite Non Lisse Pour l'Apprentissage Automatique et l'Optimisation*.
2022	Signal & Communications Group Seminar at ENSEEIHT Toulouse, France - *A Stochastic Bregman Primal-Dual Splitting Algorithm for Composite Optimization*.
2021	Thirty-fifth Conference on Neural Information Processing Systems (NeurIPS, virtual) - *Nonsmooth Implicit Differentiation for Machine Learning*.
2021	Journée du MADS Toulouse, France - *Nonsmooth Implicit Differentiation for Machine Learning*.
2021	University of Tubingen MOP Research Seminar (virtual) - *A Stochastic Bregman Primal-Dual Splitting Algorithm for Composite Optimization*.
2020	Journée du GREYC Caen, France - *Projection Free Methods for Nonsmooth Optimization in Machine Learning*.
2019	Cambridge Image Analysis Seminars - *Generalized Conditional Gradient with Augmented Lagrangian for Composite Minimization – Exact and Inexact Perspectives*.
2019	Genoa Summer School on Applied Harmonic Analysis and Machine Learning - *Inexact and Stochastic Generalized Conditional Gradient with Augmented Lagrangian and Proximal Step*.

Conference and Seminar Talks (Continued)

2019	GRETSI Lille - *Generalized Conditional Gradient with Augmented Lagrangian for Composite Minimization*.
2019	SPARS Toulouse - *Generalized Conditional Gradient with Augmented Lagrangian for Composite Optimization* **(Winner of Best Student Paper award)**.
2019	Institut de Mathématiques de Bordeaux, Séminaire IOP - *Generalized Conditional Gradient with Augmented Lagrangian for Composite Minimization*.
2019	Normastic Rouen - *Generalized Conditional Gradient with Augmented Lagrangian for Composite Minimization*.
2017	San Diego State University Student Research Symposium - *Empirical Wavelet Frames for Signal Processing*.
2015	MAA Golden Section Student Poster Session - *Comparison of Bayesian Credible Intervals to Frequentist Confidence Intervals*.
2015	Northern California Undergraduate Mathematics Conference - *Comparison of Bayesian Credible Intervals to Frequentist Confidence Intervals*.
2013	Northern California Undergraduate Mathematics Conference - *An Application of Bayesian Inference*.